

Mathematical and Logical Foundations of Computer Science

Predicate Logic (Equivalences continued)

Vincent Rahli

(some slides were adapted from Rajesh Chitnis' slides)

University of Birmingham

Where are we?

- ▶ Symbolic logic
- ▶ Propositional logic
- ▶ **Predicate logic**

Today

Equivalences:

- ▶ in Natural Deduction
- ▶ rewriting using “known” equivalences
- ▶ using semantics

Further reading:

- ▶ Chapter 8 of
http://leanprover.github.io/logic_and_proof/

Recap: Syntax

The syntax of predicate logic is defined by the following grammar:

$$t ::= x \mid f(t, \dots, t)$$

$$P ::= p(t, \dots, t) \mid \neg P \mid P \wedge P \mid P \vee P \mid P \rightarrow P \mid \forall x.P \mid \exists x.P$$

where:

- ▶ x ranges over variables
- ▶ f ranges over function symbols
- ▶ $f(t_1, \dots, t_n)$ is a well-formed term only if f has arity n
- ▶ p ranges over predicate symbols
- ▶ $p(t_1, \dots, t_n)$ is a well-formed formula only if p has arity n

The pair of a collection of function symbols, and a collection of predicate symbols, along with their arities, is called a **signature**.

The scope of a quantifier extends as far right as possible. E.g., $P \wedge \forall x.p(x) \vee q(x)$ is read as $P \wedge \forall x.(p(x) \vee q(x))$

Recap: Substitution

Substitution is defined recursively on terms and formulas:
 $P[x \backslash t]$ substitute all the free occurrences of x in P with t .

$$\begin{array}{lcl} x[x \backslash t] & = & t \\ x[y \backslash t] & = & x \\ (f(t_1, \dots, t_n))[x \backslash t] & = & f(t_1[x \backslash t], \dots, t_n[x \backslash t]) \\ (p(t_1, \dots, t_n))[x \backslash t] & = & p(t_1[x \backslash t], \dots, t_n[x \backslash t]) \\ \hline (\neg P)[x \backslash t] & = & \neg P[x \backslash t] \\ (P_1 \wedge P_2)[x \backslash t] & = & P_1[x \backslash t] \wedge P_2[x \backslash t] \\ (P_1 \vee P_2)[x \backslash t] & = & P_1[x \backslash t] \vee P_2[x \backslash t] \\ (P_1 \rightarrow P_2)[x \backslash t] & = & P_1[x \backslash t] \rightarrow P_2[x \backslash t] \\ \hline (\forall x. P)[x \backslash t] & = & \forall x. P \\ (\exists x. P)[x \backslash t] & = & \exists x. P \\ (\forall y. P)[x \backslash t] & = & \forall y. P[x \backslash t], \text{ if } y \notin \text{fv}(t) \\ (\exists y. P)[x \backslash t] & = & \exists y. P[x \backslash t], \text{ if } y \notin \text{fv}(t) \end{array}$$

The additional **conditions** ensure that **free variables do not get captured**.

These conditions can always be met by silently renaming bound variables before substituting.

Recap: $\forall$ & $\exists$ elimination and introduction rules

Natural Deduction rules for quantifiers:

$$\begin{array}{c}
 \frac{P[x \backslash y]}{\forall x.P} \quad [\forall I] \qquad \frac{\forall x.P}{P[x \backslash t]} \quad [\forall E] \qquad \frac{P[x \backslash t]}{\exists x.P} \quad [\exists I] \qquad \frac{\exists x.P \quad \begin{array}{c} \overline{P[x \backslash y]}^1 \\ \vdots \\ Q \end{array}}{Q}^1 \quad [\exists E]
 \end{array}$$

Condition:

- ▶ for $[\forall I]$: y must not be free in any not-yet-discharged hypothesis or in $\forall x.P$
- ▶ for $[\forall E]$: $\mathbf{fv}(t)$ must not clash with $\mathbf{bv}(P)$
- ▶ for $[\exists I]$: $\mathbf{fv}(t)$ must not clash with $\mathbf{bv}(P)$
- ▶ for $[\exists E]$: y must not be free in Q or in not-yet-discharged hypotheses or in $\exists x.P$

Recap: Example of a proof

here is a proof of $(\forall z.p(z)) \rightarrow \forall x.p(x) \vee q(x)$.

$$\frac{\frac{\frac{\overline{\quad} 1}{\forall z.p(z)} [\forall E]}{p(y)} [\vee I_L]}{\frac{p(y) \vee q(y)}{\forall x.p(x) \vee q(x)} [\forall I]} 1 [\rightarrow I]$$

Conditions:

- ▶ y does not occur free in not-yet-discharged hypotheses or in $\forall x.p(x) \vee q(x)$
- ▶ y does not clash with bound variables in $p(z)$

Recap: Models

Models: a model provides the interpretation of all symbols

Given a **signature** $\langle\langle f_1^{k_1}, \dots, f_n^{k_n} \rangle, \langle p_1^{j_1}, \dots, p_m^{j_m} \rangle\rangle$

- ▶ of function symbols f_i of arity k_i , for $1 \leq i \leq n$
- ▶ of predicate symbols p_i of arity j_i , for $1 \leq i \leq m$

a **model** is a structure $\langle D, \langle \mathcal{F}_{f_1}, \dots, \mathcal{F}_{f_n} \rangle, \langle \mathcal{R}_{p_1}, \dots, \mathcal{R}_{p_m} \rangle \rangle$

- ▶ of a non-empty domain D
- ▶ interpretations $\mathcal{F}_{f_i}$ for function symbols f_i
- ▶ interpretations $\mathcal{R}_{p_i}$ for predicate symbols p_i

Models of predicate logic replace **truth assignments** for propositional logic

Variable valuations:

- ▶ a partial function v
- ▶ that map variables to D
- ▶ i.e., a mapping of the form $x_1 \mapsto d_1, \dots, x_n \mapsto d_n$

Recap: Semantics of Predicate Logic

Given a **model** M with domain D and a **variable valuation** v :

- ▶ $\llbracket t \rrbracket_v^M$ gives meaning to the term t w.r.t. M and v
- ▶ $\models_{M,v} P$ gives meaning to the formula P w.r.t. M and v

Meaning of terms:

- ▶ $\llbracket x \rrbracket_v^M = v(x)$
- ▶ $\llbracket f(t_1, \dots, t_n) \rrbracket_v^M = \mathcal{F}_f(\langle \llbracket t_1 \rrbracket_v^M, \dots, \llbracket t_n \rrbracket_v^M \rangle)$

Meaning of formulas:

- ▶ $\models_{M,v} p(t_1, \dots, t_n)$ iff $\langle \llbracket t_1 \rrbracket_v^M, \dots, \llbracket t_n \rrbracket_v^M \rangle \in \mathcal{R}_p$
- ▶ $\models_{M,v} \neg P$ iff $\not\models_{M,v} P$
- ▶ $\models_{M,v} P \wedge Q$ iff $\models_{M,v} P$ and $\models_{M,v} Q$
- ▶ $\models_{M,v} P \vee Q$ iff $\models_{M,v} P$ or $\models_{M,v} Q$
- ▶ $\models_{M,v} P \rightarrow Q$ iff $\models_{M,v} Q$ whenever $\models_{M,v} P$
- ▶ $\models_{M,v} \forall x. P$ iff for every $d \in D$ we have $\models_{M,(v,x \mapsto d)} P$
- ▶ $\models_{M,v} \exists x. P$ iff there exists a $d \in D$ such that $\models_{M,(v,x \mapsto d)} P$

Recap: Logical equivalences for Propositional Logic

The same equivalences hold as in Propositional Logic:

- ▶ De Morgan's law (I): $\neg(A \vee B) \leftrightarrow (\neg A \wedge \neg B)$
- ▶ De Morgan's law (II): $\neg(A \wedge B) \leftrightarrow (\neg A \vee \neg B)$
- ▶ Implication elimination: $(A \rightarrow B) \leftrightarrow (\neg A \vee B)$
- ▶ Commutativity of $\wedge$: $(A \wedge B) \leftrightarrow (B \wedge A)$
- ▶ Commutativity of $\vee$: $(A \vee B) \leftrightarrow (B \vee A)$
- ▶ Associativity of $\wedge$: $((A \wedge B) \wedge C) \leftrightarrow (A \wedge (B \wedge C))$
- ▶ Associativity of $\vee$: $((A \vee B) \vee C) \leftrightarrow (A \vee (B \vee C))$
- ▶ Distributivity of $\wedge$ over $\vee$: $(A \wedge (B \vee C)) \leftrightarrow ((A \wedge B) \vee (A \wedge C))$
- ▶ Distributivity of $\vee$ over $\wedge$: $(A \vee (B \wedge C)) \leftrightarrow ((A \vee B) \wedge (A \vee C))$
- ▶ Double negation elimination: $(\neg\neg A) \leftrightarrow A$
- ▶ Idempotence: $(A \wedge A) \leftrightarrow A$ and $(A \vee A) \leftrightarrow A$

Recap: Logical Equivalences

In addition, the following hold (some hold only classically):

- ▶ $(\forall x.A \wedge B) \leftrightarrow ((\forall x.A) \wedge (\forall x.B))$
- ▶ $(\exists x.A \vee B) \leftrightarrow ((\exists x.A) \vee (\exists x.B))$
- ▶ $(\neg \forall x.A) \leftrightarrow (\exists x. \neg A)$
- ▶ $(\neg \exists x.A) \leftrightarrow (\forall x. \neg A)$
- ▶ $(\forall x.A) \leftrightarrow A$ if $x \notin \text{fv}(A)$
- ▶ $(\exists x.A) \leftrightarrow A$ if $x \notin \text{fv}(A)$
- ▶ $(\forall x.A \vee B) \leftrightarrow ((\forall x.A) \vee B)$ if $x \notin \text{fv}(B)$
- ▶ $(\exists x.A \wedge B) \leftrightarrow ((\exists x.A) \wedge B)$ if $x \notin \text{fv}(B)$
- ▶ $(\forall x.A \rightarrow B) \leftrightarrow ((\exists x.A) \rightarrow B)$ if $x \notin \text{fv}(B)$
- ▶ $(\exists x.A \rightarrow B) \leftrightarrow ((\forall x.A) \rightarrow B)$ if $x \notin \text{fv}(B)$
- ▶ $(\forall x.A \rightarrow B) \leftrightarrow (A \rightarrow \forall x.B)$ if $x \notin \text{fv}(A)$
- ▶ $(\exists x.A \rightarrow B) \leftrightarrow (A \rightarrow \exists x.B)$ if $x \notin \text{fv}(A)$

Recap: Logical Equivalences

As before: if $(P \leftrightarrow Q \text{ or } Q \leftrightarrow P)$ and P occurs in A , then replacing P by Q in A leads to a formula B , such that $A \leftrightarrow B$

Also,

Semantical equivalence: two formulas P and Q are equivalent if for all models M and valuations v , $\models_{M,v} P$ iff $\models_{M,v} Q$

Logical Equivalences

As before to prove a logical equivalence $A \leftrightarrow B$, we will prove:

- ▶ that we can derive B from A
- ▶ that we can derive A from B

We will start by proving:

- ▶ $(\forall x.A) \leftrightarrow A$ if $x \notin \text{fv}(A)$
- ▶ $(\exists x.A) \leftrightarrow A$ if $x \notin \text{fv}(A)$
- ▶ $(\forall x.A \vee B) \leftrightarrow ((\forall x.A) \vee B)$ if $x \notin \text{fv}(B)$
- ▶ $(\exists x.A \wedge B) \leftrightarrow ((\exists x.A) \wedge B)$ if $x \notin \text{fv}(B)$

We will use the following result:

Lemma (L1): if $x \notin \text{fv}(A)$ then $A[x \backslash t] = A$

Logical Equivalences

Prove $(\forall x.A) \leftrightarrow A$ if $x \notin \text{fv}(A)$ in Natural Deduction

Here is a proof of the right-to-left implication (constructive):

$$\frac{A[x \backslash y]A}{\forall x.A} \quad [\forall I]$$

- ▶ pick y such that it does not occur in A
- ▶ by L1, because $x \notin \text{fv}(A)$ then $A[x \backslash y] = A$

Here is a proof of the left-to-right implication (constructive):

$$\frac{\forall x.A}{AA[x \backslash y]} \quad [\forall E]$$

- ▶ by L1, because $x \notin \text{fv}(A)$ then $A[x \backslash y] = A$
- ▶ pick y such that it does not occur in A

Logical Equivalences

Prove $(\exists x.A) \leftrightarrow A$ if $x \notin \text{fv}(A)$ in Natural Deduction

Here is a proof of the right-to-left implication (constructive):

$$\frac{A[x \backslash y]A}{\exists x.A} [\exists I]$$

- ▶ pick y such that it does not occur in A
- ▶ by L1, because $x \notin \text{fv}(A)$ then $A[x \backslash y] = A$

Here is a proof of the left-to-right implication (constructive):

$$\frac{\exists x.A \quad \overline{AA[x \backslash y]}}{A} 1 [\exists E]$$

- ▶ by L1, because $x \notin \text{fv}(A)$ then $A[x \backslash y] = A$
- ▶ pick y such that it does not occur in A

Logical Equivalences

Prove that $(\forall x.A \vee B) \leftrightarrow ((\forall x.A) \vee B)$ if $x \notin \text{fv}(B)$ in Natural Deduction

Here is a proof of the left-to-right implication (classical):

$$\frac{\frac{\frac{}{B \vee \neg B} \quad [LEM] \quad \frac{\frac{\frac{}{B} \quad 1}{(\forall x.A) \vee B} [\vee I_R]}{B \rightarrow (\forall x.A) \vee B} 1 \quad [\rightarrow I] \quad \frac{\frac{\frac{}{\neg B} \quad 2}{\neg B \rightarrow (\forall x.A) \vee B} [\vee I_L]}{\neg B \rightarrow (\forall x.A) \vee B} 2 \quad [\rightarrow I]}{(\forall x.A) \vee B} [\vee E]$$

where Π is:

$$\frac{\frac{\frac{}{\forall x.A \vee B} \quad [\vee E] \quad \frac{\frac{}{A[x \setminus y]} \quad 3}{A[x \setminus y] \rightarrow A[x \setminus y]} 3 \quad [\rightarrow I] \quad \frac{\frac{\frac{}{\neg B} \quad 2 \quad \frac{}{B} \quad 4}{\perp} [\neg E]}{A[x \setminus y]} [\perp E]}{B \rightarrow A[x \setminus y]} 4 \quad [\rightarrow I]}{A[x \setminus y]} [\vee E] \quad \frac{A[x \setminus y]}{\forall x.A} [\forall I]$$

- ▶ pick y such that it does not occur in A or B
- ▶ by L1, because $x \notin \text{fv}(B)$ then $B[x \setminus y] = B$

Logical Equivalences

Prove that $(\forall x.A \vee B) \leftrightarrow ((\forall x.A) \vee B)$ if $x \notin \text{fv}(B)$ in Natural Deduction

Here is a proof of the right-to-left implication (constructive):

$$\begin{array}{c}
 \frac{\frac{\frac{}{1} \quad \forall x.A}{A[x \backslash y]} [\forall E]}{A[x \backslash y] \vee B} [\vee I_L] \quad \frac{\frac{}{2} \quad B}{A[x \backslash y] \vee B} [\vee I_R]}{\frac{(\forall x.A) \vee B \quad \frac{(\forall x.A) \rightarrow A[x \backslash y] \vee B}{B \rightarrow A[x \backslash y] \vee B} \quad 1 \ [\rightarrow I] \quad 2 \ [\rightarrow I]}{A[x \backslash y] \vee B} [\vee E]} \\
 \frac{}{\forall x.A \vee B} [\forall I]
 \end{array}$$

- ▶ pick y such that it does not occur in A or B
- ▶ by L1, because $x \notin \text{fv}(B)$ then $B[x \backslash y] = B$

Logical Equivalences

Prove that $(\exists x.A \wedge B) \leftrightarrow ((\exists x.A) \wedge B)$ if $x \notin \text{fv}(B)$ in Natural Deduction

Here is a proof of the left-to-right implication (constructive):

$$\begin{array}{c}
 \frac{}{A[x \backslash y] \wedge B} 1 \\
 \frac{}{A[x \backslash y]} [\wedge E_L] \quad \frac{}{A[x \backslash y] \wedge B} 1 \\
 \frac{}{\exists x.A} [\exists I] \quad \frac{}{B} [\wedge E_R] \\
 \frac{}{(\exists x.A) \wedge B} [\wedge I] \\
 \frac{}{\exists x.A \wedge B} [\exists E]
 \end{array}$$

- ▶ pick y such that it does not occur in A or B
- ▶ by L1, because $x \notin \text{fv}(B)$ then $B[x \backslash y] = B$

Logical Equivalences

Prove that $(\exists x.A \wedge B) \leftrightarrow ((\exists x.A) \wedge B)$ if $x \notin \text{fv}(B)$ in Natural Deduction

Here is a proof of the right-to-left implication (constructive):

$$\begin{array}{c}
 \dfrac{\dfrac{\dfrac{(\exists x.A) \wedge B}{\exists x.A} [\wedge E_L] \quad \dfrac{\dfrac{\dfrac{A[x \backslash y]}{A[x \backslash y]} 1 \quad \dfrac{(\exists x.A) \wedge B}{B} [\wedge E_R]}{A[x \backslash y] \wedge B} [\wedge I]}{\dfrac{A[x \backslash y] \wedge B}{\exists x.A \wedge B} [\exists I]} [\exists E]} \exists x.A \wedge B
 \end{array}$$

- ▶ pick y such that it does not occur in A or B
- ▶ by L1, because $x \notin \text{fv}(B)$ then $B[x \backslash y] = B$

Logical Equivalences

We will now prove the following using the other equivalences:

- ▶ $(\forall x.A \rightarrow B) \leftrightarrow ((\exists x.A) \rightarrow B)$ if $x \notin \text{fv}(B)$
- ▶ $(\exists x.A \rightarrow B) \leftrightarrow ((\forall x.A) \rightarrow B)$ if $x \notin \text{fv}(B)$

Prove that $(\forall x.A \rightarrow B) \leftrightarrow ((\exists x.A) \rightarrow B)$ if $x \notin \text{fv}(B)$ using the other equivalences

- ▶ $\forall x.A \rightarrow B$
- ▶ $\leftrightarrow \forall x.\neg A \vee B$ – using implication elimination
- ▶ $\leftrightarrow (\forall x.\neg A) \vee B$ – using $(\forall x.A \vee B) \leftrightarrow ((\forall x.A) \vee B)$ if $x \notin \text{fv}(B)$
- ▶ $\leftrightarrow (\neg \exists x.A) \vee B$ – using $(\neg \exists x.A) \leftrightarrow (\forall x.\neg A)$
- ▶ $\leftrightarrow (\exists x.A) \rightarrow B$ – using implication elimination

Logical Equivalences

Prove that $(\exists x.A \rightarrow B) \leftrightarrow ((\forall x.A) \rightarrow B)$ if $x \notin \text{fv}(B)$ using the other equivalences

- ▶ $\exists x.A \rightarrow B$
- ▶ $\leftrightarrow \exists x.\neg A \vee B$ – using implication elimination
- ▶ $\leftrightarrow (\exists x.\neg A) \vee (\exists x.B)$ – using $(\exists x.A \vee B) \leftrightarrow ((\exists x.A) \vee (\exists x.B))$
- ▶ $\leftrightarrow (\exists x.\neg A) \vee B$ – using $(\exists x.A) \leftrightarrow A$ if $x \notin \text{fv}(A)$
- ▶ $\leftrightarrow (\neg \forall x.A) \vee B$ – using $(\neg \forall x.A) \leftrightarrow (\exists x.\neg A)$
- ▶ $\leftrightarrow (\forall x.A) \rightarrow B$ – using implication elimination

Logical Equivalences

We will now prove the following using semantics:

- ▶ $(\forall x.A \rightarrow B) \leftrightarrow (A \rightarrow \forall x.B)$ if $x \notin \text{fv}(A)$
- ▶ $(\exists x.A \rightarrow B) \leftrightarrow (A \rightarrow \exists x.B)$ if $x \notin \text{fv}(A)$

We will use following result:

Lemma (L2): if $x \notin \text{fv}(A)$, then $\models_{M,v,x \mapsto d} A$ iff $\models_{M,v} A$

Logical Equivalences

Prove $(\forall x.A \rightarrow B) \leftrightarrow (A \rightarrow \forall x.B)$ if $x \notin \text{fv}(A)$ using the semantics method

Assume $x \notin \text{fv}(A)$, M is a model with domain D and v a valuation
Left-to-right implication:

- ▶ if $\models_{M,v} \forall x.A \rightarrow B$ then $\models_{M,v} A \rightarrow \forall x.B$
 - ▶ to prove: $\models_{M,v} A \rightarrow \forall x.B$, i.e., $\models_{M,v} \forall x.B$ whenever $\models_{M,v} A$
 - ▶ assume $\models_{M,v} A$ and prove $\models_{M,v} \forall x.B$, i.e., for all $d \in D$, $\models_{M,v,x \mapsto d} B$
 - ▶ assumption: $\models_{M,v} \forall x.A \rightarrow B$, i.e., for all $e \in D$, $\models_{M,v,x \mapsto e} B$ whenever $\models_{M,v,x \mapsto e} A$
 - ▶ because $\models_{M,v} A$ by L2, $\models_{M,v,x \mapsto d} A$
 - ▶ instantiating this assumption with d gives us: $\models_{M,v,x \mapsto d} B$ whenever $\models_{M,v,x \mapsto d} A$
 - ▶ therefore, because $\models_{M,v,x \mapsto d} A$ is true, $\models_{M,v,x \mapsto d} B$ is also true

Logical Equivalences

Right-to-left implication:

- ▶ if $\models_{M,v} A \rightarrow \forall x.B$ then $\models_{M,v} \forall x.A \rightarrow B$
 - ▶ to prove: $\models_{M,v} \forall x.A \rightarrow B$, i.e., for all $d \in D$, $\models_{M,v,x \mapsto d} B$ whenever $\models_{M,v,x \mapsto d} A$
 - ▶ assume $d \in D$ and $\models_{M,v,x \mapsto d} A$, and prove $\models_{M,v,x \mapsto d} B$
 - ▶ by L2, we can assume $\models_{M,v} A$
 - ▶ assumption: $\models_{M,v} A \rightarrow \forall x.B$, i.e., $\models_{M,v} \forall x.B$ whenever $\models_{M,v} A$
 - ▶ because $\models_{M,v} A$, we can assume $\models_{M,v} \forall x.B$, i.e., for all $e \in D$, $\models_{M,v,x \mapsto e} B$
 - ▶ instantiating this assumption using d , we get to assume $\models_{M,v,x \mapsto d} B$, which is what we wanted to prove

Conclusion

What did we cover today?

- ▶ Equivalence using Natural Deduction
- ▶ Rewriting using “known” equivalences
- ▶ Equivalences using semantics

Further reading:

- ▶ Chapter 8 of
http://leanprover.github.io/logic_and_proof/

Next time?

- ▶ Theorem Proving